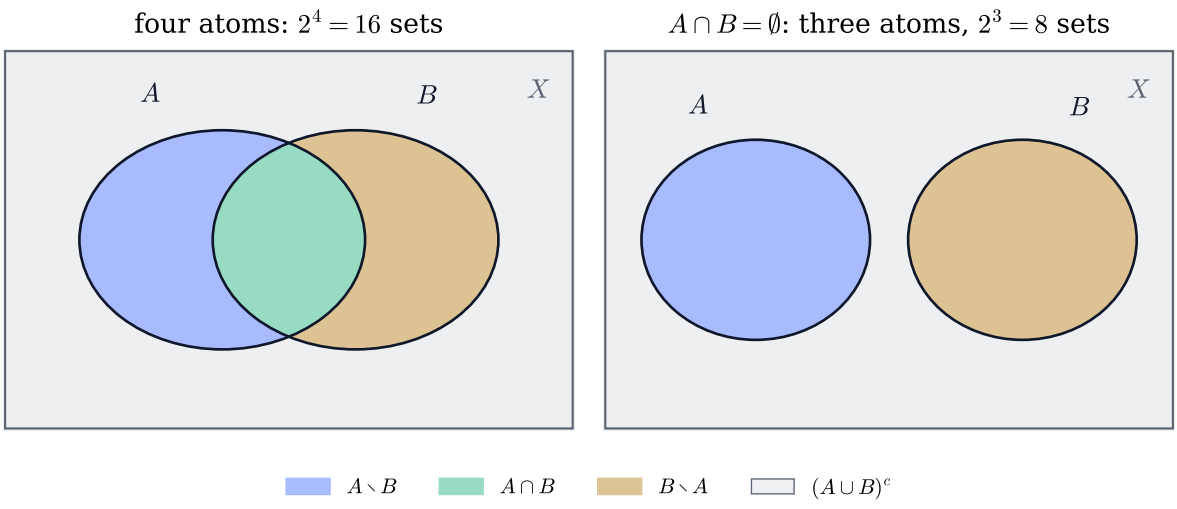
four atoms: $2^4 = 16$ sets
A
B
X
$A \cap B = \emptyset$: three atoms, $2^3 = 8$ sets
A
B
X
$A \setminus B$
$A \cap B$
$B \setminus A$
$(A \cup B)^c$